

TOPIC 19/60

How are LLMs Trained?

The journey from raw text
to reasoning assistant.

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THE LIFECYCLE

Training an LLM isn't just one step. It's a structured, three-stage evolution.

1. Pre-training

Learning patterns from the internet.

2. Supervised Fine-Tuning

Learning to follow instructions.

3. RLHF

Polishing & Human Alignment.

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STAGE 1: PRE-TRAINING



The "School" Phase

The goal here is simple: **Next-Token Prediction**. Read the entire internet and guess the next word.

By doing this billions of times, the model learns grammar, facts, reasoning, and even coding, purely by statistical exposure.

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THE VOLUME

Massive Compute

This is the expensive part. We aren't talking about reading a few books.

The Cost:

Tens of thousands of H100 GPUs running for months.
Consuming the energy of a small city to turn electricity into "world knowledge."

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STAGE 2

SFT

Instruction Tuning

A raw pre-trained model is just a "text completer." If you ask it a question, it might just ask another question.

SFT: We feed it thousands of high-quality examples: "Prompt -> Answer." It learns how to be an assistant, not just a document completer.

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STAGE 3

RLHF

Human Feedback

How do we know if the answer is "good"? We ask humans.

The Process:

1. Model generates multiple answers.
2. Human ranks them (A is better than B).
- > 3. A Reward Model is trained to mimic these human preferences.

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WHY ALIGNMENT MATTERS

SAFETY

Refusal to be toxic.

Ensuring the model doesn't generate illegal or harmful content.

UTILITY

Being Helpful.

Prioritizing concise, accurate, and structured answers over "creative" nonsense.

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FINAL STEPS

Post-Training

Before the model reaches your computer, it gets "compressed" and tuned for efficiency.

Quantization: Shrinking the model size (e.g., from 16-bit to 4-bit) so it can run on smaller hardware with minimal accuracy loss.

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THE FINAL TEST

Benchmarks

Evaluating against standard exams (MMLU, HumanEval) to measure logic, coding, and knowledge.

Red Teaming

Attempting to trick or "break" the model to find safety holes before release.

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CONCEPT UNLOCKED

- 1 Pre-training: Patterns
- 2 Fine-Tuning: Instructions
- 3 RLHF: Human Feedback
- 🌟 Final Polish: Quantization

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